A. Hash of digits

Time limit: 1.5 s · Memory limit: 256 MB

You have N strings s^i consisting of digits and hashes h^i of these strings, where hash of string s^i equal to h^i. But one day You noticed that hashes and strings were mixed. You need to determine which of the numbers are hashes and which are strings consisting of digits. Also map each string to its hash. Hash should be calculated by formula below. All operations must be performed modulo $1e9 + 7$:

$$h = (\sum_{i=0}^{|s|-1}(s_i - 47) \times 11^i) \mod (10^9 + 7)$$

Input

Given the number N - number of strings ($1 \le N \le 10^4$). Following $2 \cdot N$ lines consist from strings and hashes. The length of each string does not exceed 100. It is guaranteed that for each string there is a corresponding hash.

Output

In each line print string with corresponding hash like in example below in order their input.

Note

In first example, hash of string //111// was calculated as follows:

$$h = \Big(\big(((49 - 47) \times 11^0) \mod (1e9 + 7) + ((49 - 47) \times 11^1) \mod (1e9 + 7)\big) \mod (1e9 + 7) +$$

$$+((49 - 47) \times 11^2) \mod (1e9 + 7)\Big) \mod (1e9 + 7) = 2 + 22 + 242 = 266$$

Note, that the resulting number should be positive.

Examples

Input 1

```
3
334368200
111
100000
266
123456789
93085
```

Output 1

```
Hash of string "111" is 266
Hash of string "123456789" is 334368200
Hash of string "93085" is 100000
```

Input 2

```
3
13903
338308044731467504464331 3
9839
813695185
425675346
3461762860035486
```

Output 2

```
Hash of string "338308044731467504464331 3" is
425675346
Hash of string "9839" is 13903
Hash of string "3461762860035486" is 813695185
```

B. Plagiarism

Time limit: 1 s · Memory limit: 256 MB

Alimzhan wanted to check his students for cheating and he added the parasite word to his exam. He believes that students cheat if the parasite word occurs in the same place in two different results of two students. Alimzhan heard that two of his most beloved students were copied from each other. Help Alimzhan know how many times the parasite word is repeated in the same position in his beloved students' results.

Input

The first line contains string s_1 ($1 \le |s_1| \le 4 \cdot 10^5$), consisting of lowercase Latin letters — the result of the first student.

The second line contains string s_2 ($1 \le |s_2| \le 4 \cdot 10^5$), consisting of lowercase Latin letters — the result of the second student.

The third line contains string t ($1 \le |t| \le 4 \cdot 10^5$), consisting of lowercase Latin letters — the parasite word.

Output

Print one integer — the number of positions i such that the parasite word t occurs in s_1 starting at position i and occurs in s_2 starting at the same position i.

Note

The strings are long, so comparing t with every position directly is too slow. Compute the hashes of all substrings of the needed length with the Rabin–Karp technique (prefix hashes and one precomputed power table) and compare hashes instead of strings.

A single hash can collide on adversarial data; a pair of hashes with two different moduli is the usual remedy.

Examples

Input 1

```
kbtuutbkkbtu
utbkkbtukbtu
kbtu
```

Output 1

```
1
```

Input 2

```
aaaaa
aaaaa
a
```

Output 2

```
5
```

Input 3

```
abracadabra
abacabaabac
ab
```

Output 3

```
2
```

C. Dominating Patterns

Time limit: 1 s · Memory limit: 256 MB

The archaeologists are going to decipher a very mysterious "language". Now, they know many language patterns. Each pattern can be treated as a string of lowercase Latin letters. There is a large text they want to analyze. What matters most is that which patterns are the *dominating patterns. Frequency* of the pattern is the number of its appearances in the text as a substring. Pattern is dominating if the frequency the pattern is is not less than the frequency of any other pattern. It is your job to find all dominating patterns.

Input

The input consists of multiple test cases.

The first line of each test case is an integer N ($1 \leq N \leq 150$) - the number of patterns. Each of the following N lines contain single string of length in range $[1, 70]$ - patterns. The next line contains a single string of length in range $[1, 10^5]$ - text.

At the end of the input file there is a 0 indicating the end of an input file.

Output

For each test case, output the dominating patterns. If there are several dominating patterns, output them in order they appear in the input.

Note

In the first test case, the frequency of pattern aba is 2, and of pattern bac is 1.

In the second test case, the frequency of patterns $alpha$ and $haha$ is 2, and all other patterns has frequency not exceeding 1.

Examples

Input 1

```
2
aba
bab
abababababac
6
beta
alpha
haha
delta
dede
tata
dedeltalphahahahototatalpha
0
```

Output 1

```
4
aba
2
alpha
haha
```

D. Restore the string

Time limit: 1 s · Memory limit: 256 MB

Jonathan tried to write Rabin-Karp algorithm. He had a string S and he used the following formula to calculate hash of string:

$$\sum_{i=0}^{|S|-1} (S_i - 97) \cdot 2^i$$

Thus, he has written hashes of all prefixes of string S. Unfortunately, Jonathan forgot his string S. So, he asked your help in restoring this string.

Input

The first line of the input contains the only integer N - the length of string S ($1 \leqslant N \leqslant 50$).

The second line contains N integers p_i - hashes of all prefixes of string S.

It is guaranteed that each hash does not exceed $2 \cdot 10^{18}$.

Output

Print the string S.

Note

Use larger integer data structures (like "long long" In C++).

Examples

Input 1

```
5
7 15 59 147 371
```

Output 1

```
hello
```

Input 2

```
5
22 50 118 206 254
```

Output 2

```
world
```

E. Distinct substrings

Time limit: 4 s · Memory limit: 256 MB

Given string s. How many different substrings it contains?

Input

The only line contains string s ($0 \leq |s| \leq 2000$), consisting of lowercase Latin letters. The string may be empty.

Output

Print one integer — the number of distinct substrings of s. The empty substring is not counted.

Note

You can solve this problem using hashes. Naive $O(N^3)$ solutions must fail.

Examples

Input 1	Output 1
aab	5

Input 2	Output 2
abcde	15

F. Substrings

Time limit: 3 s · Memory limit: 512 MB

You have string S and q queries. Each query given a pair of integers L and R; the answer for a query is number of times substring from L to R appears in S as substring.

Input

In first line given string S $(1 \leq |s| \leq 2000)$.

In second line given integer q $(1 \leq q \leq 5 * 10^4)$.

Then next q lines contains queries, each query is two integer L and R $(1 \leq L \leq R \leq |s|)$, substring you must to check.

Output

Print q lines with one single integer, answer for each query.

Examples

Input 1

```
abracadabra
5
1 1
1 2
3 4
1 5
1 4
```

Output 1

```
5
2
2
1
2
```